SDC Hackathon 2026 | Domain 1 · Future-Ready Workforce | SDG 4 Quality Education

Calador – AI Career & Skills Navigator



From “who I am today” to “where I can be in the AI era” — in minutes, with every step backed by a cited source.

- Profile
- Future Skills Radar
- Role match
- Reskilling pathway

Design By: Team Calador

Malaysia is adopting AI faster than it is training for it.

56%

of Malaysian professionals already use AI tools at work

26%

have received formal AI training

600,000+

workers need reskilling within 3–5 years



Sources: Hays Asia Salary Guide 2025; MOSTI statement — both as cited in the SDC 2026 Domain 1 brief.

Four groups face the same question: **what should we learn next?**

01
Working adults

at risk of job displacement

02
Fresh graduates & career switchers

choosing where to go next

03
HR & L&D managers

deciding which roles to train teams for

04
TVET & university career centres

advising students at scale

THREE MAIN PROBLEMS:

Unclear Skills
Workers don’t know which AI skills are worth learning.

Uncertain Careers
Employers can’t tell which roles to train for.

Outdated Pathways
Learning pathways are static lists — out of date when published.

SDG Impact

SDG 4

Quality Education

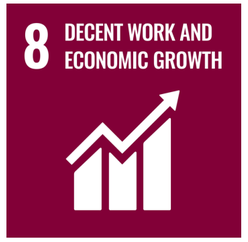


Main target: 4.4

Builds relevant skills for employment through personalised skills-gap analysis and reskilling pathways.

SDG 8

Decent Work and Economic Growth



Supporting targets: 8.2 and 8.5

Supports preparation for changing job requirements and future employment.

Insight: AI progress is unpredictable, so a one-time skills assessment is not enough. People need guidance that keeps updating.

Our Solution: Aiversecity

1

Profile

Form, CV upload and a short potential questionnaire

2

Future Skills Radar

Which skills are rising or cooling — from 3 sources

3

Ranked roles

Match score + skills gap for every future role

4

Reskilling pathway

Free resources, hours, weeks, progress & badges

Streamlit web app

Not a chatbot — no LLM

Every result cites its source

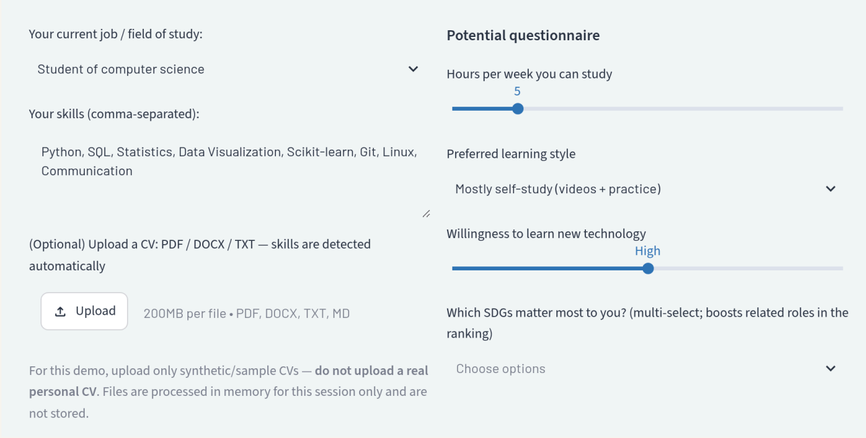
Nothing is invented

Core Features – From Skills Assessment to Future-Ready Careers

01

Skills Profile Intake

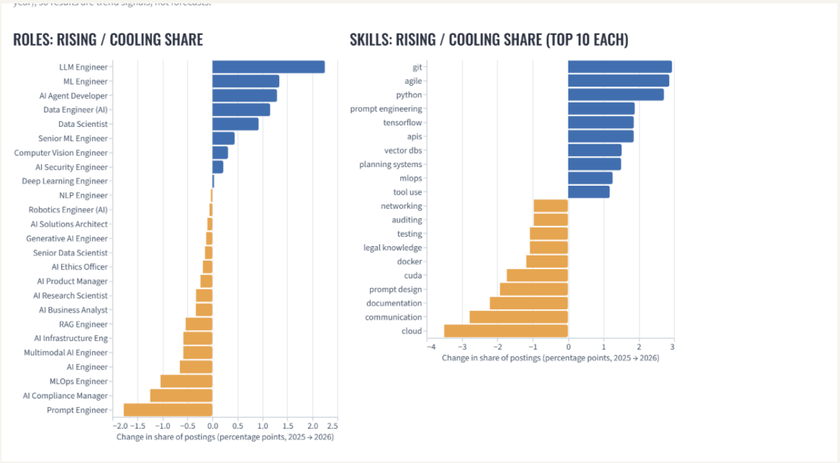
Job/ field dropdown, skills, CV upload (PDF · DOCX · TXT) with automatic skill detection, SDG interests, and a potential questionnaire.



02

Future Skills Radar

2025 vs 2026 job - posting shares + live arXiv cs.AI papers + live Arbeitnow job ads, combined into skill momentum.

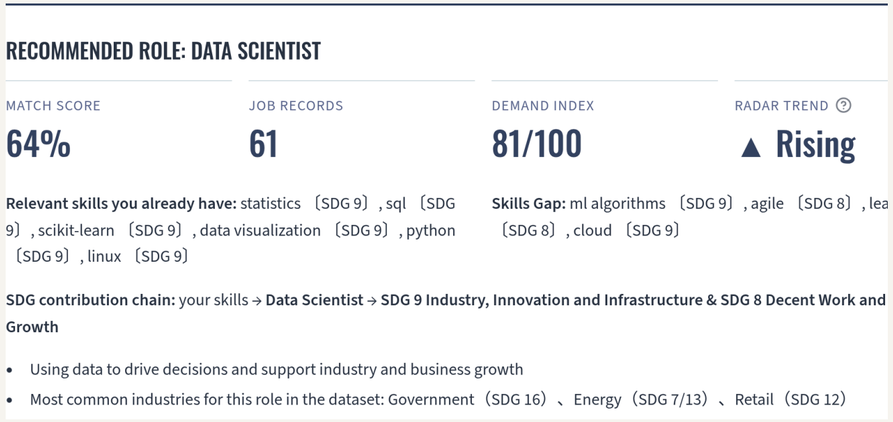


Core Features – From Skills Assessment to Future-Ready Careers

03

Future-role matching

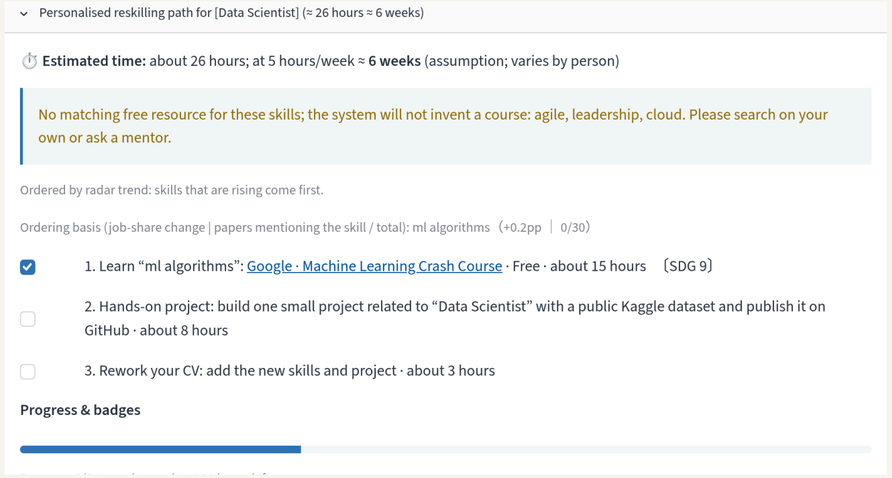
Explainable match score with matched skills, missing skills, postings, demand score and trend status.



04

Personalised reskilling pathway

Gap skills ordered by momentum, then a hands-on Kaggle → GitHub project and a CV step. Course count, order and pace adapt to study hours, learning style and willingness.

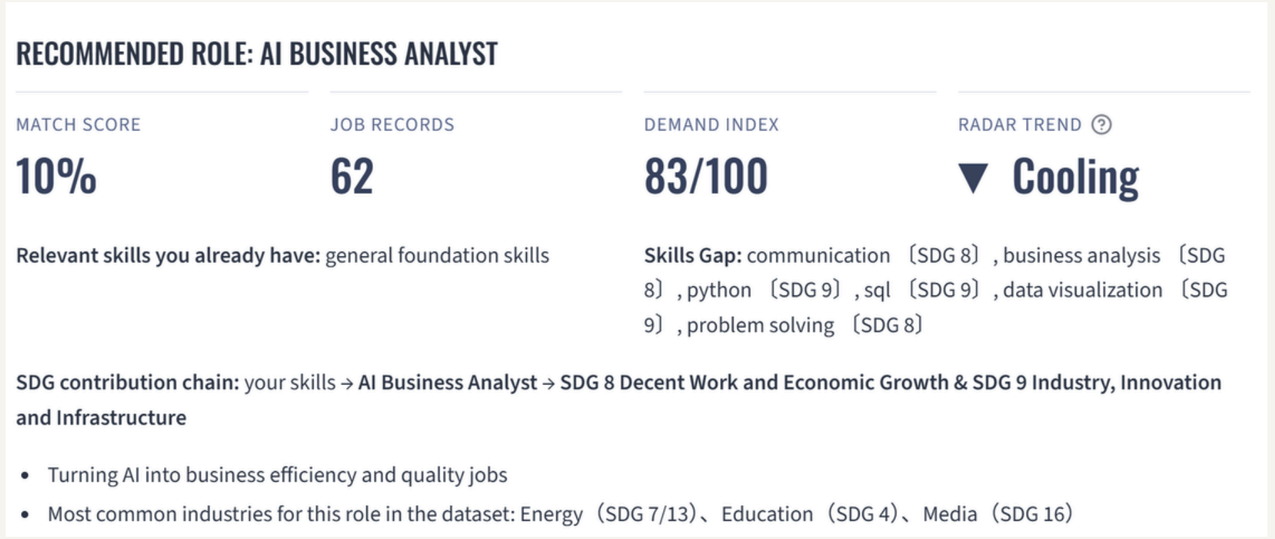


Core Features – From Skills Assessment to Future-Ready Careers

05

SDG Mapping

Every role and skill is linked to at least one of the 17 UN SDGs. Pick the goals you care about.



06

Core Features – From Skills Assessment to Future-Ready Careers

05

06

Handling Uncertainty

Slow, moderate and rapid-change scenarios with an uncertainty range, labelled as estimates.



Additional Features: Fairness, Access & Team Insights

Team / batch mode

HR and career centres upload a member CSV (template provided) to get a workforce skills-gap heatmap, regional SDG contribution and a downloadable result.

Automatic re-planning

Compares rankings with and without trend signals and flags when recommendations change.

Fairness check

Skills-only matching option and a table of scores across backgrounds with identical skills.

Live job-market panel

Skill share in live job ads vs the dataset’s 2026 share.

3 languages

English, Malay and Chinese across whole interface.

Low-bandwidth mode

Tables instead of charts, no live fetch, cached data only — usable on mobile.

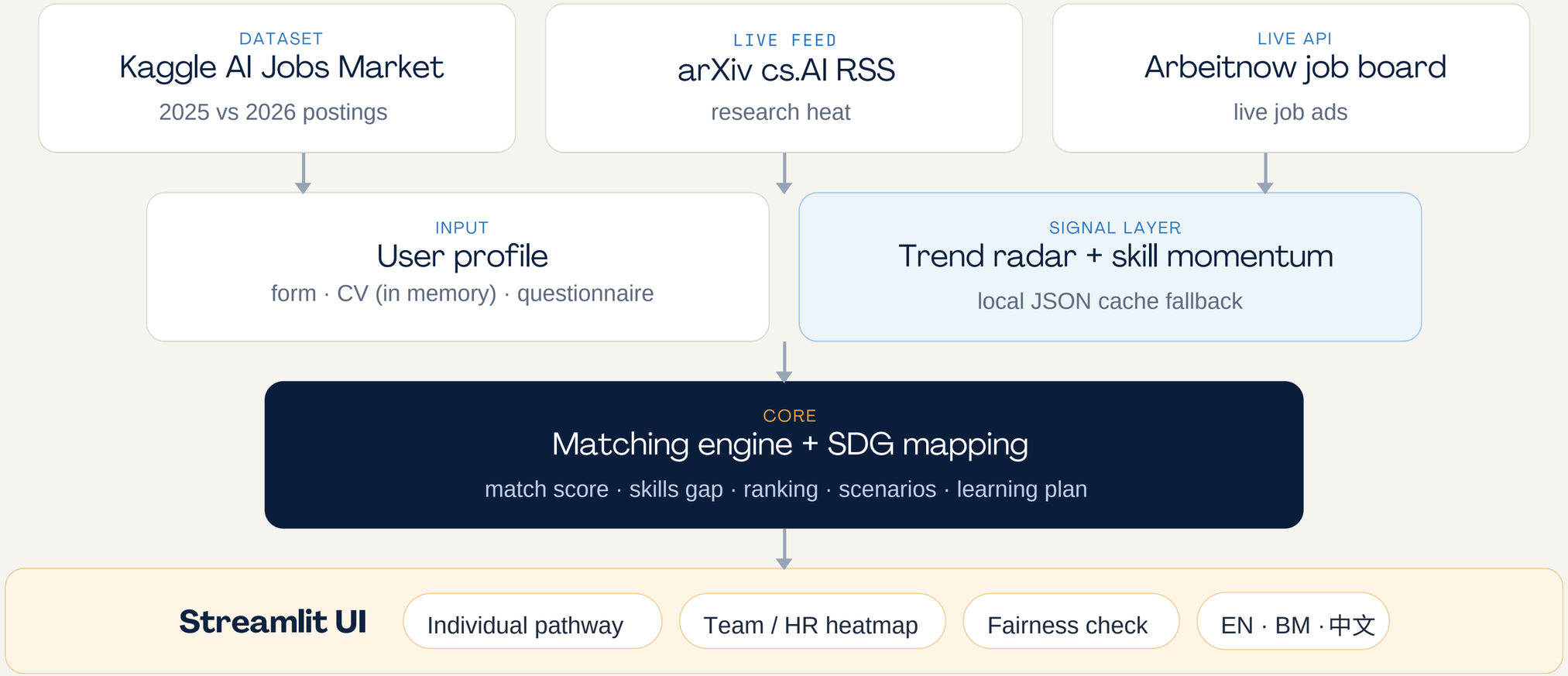
Motivation & export

Progress checkboxes, skill badges, plan download as CSV or Markdown.

Adjustable assumptions

Scenario growth multipliers and the ± uncertainty range can be changed in the sidebar.

System architecture



Inside the engine: transparent rules you can check.

MATCH SCORE (SHOWN TO THE USER)

90% × skill coverage +10% × background relevance

Background weight can be switched off for skills-only matching.

SKILL MOMENTUM (ORDERS THE RADAR & PATHWAY)

Δ job share (pp) + 0.1 × research heat % + 0.05 × live-ad share %

Skills: rising ≥ +1.0 pp, cooling ≤ −1.0 pp between 2025 and 2026 posting shares (roles: ±0.3 pp).

RANKING (ORDER ONLY — NEVER CHANGES THE MATCH SCORE)

match score + trend bonus × willingness + 5 pts per chosen SDG

Willingness scales trend weight 0.5× / 1.0× / 1.5×. Study hours set 2–4 courses (+1 at very high willingness).

All weights are team-set assumptions, shown and adjustable in the app — not model outputs.

Data Foundation and Tech Stack

1,500 AI job postings

25 future AI roles

93 distinct skills

14 countries / regions

Kaggle AI Jobs Market 2025–2026 · 624 postings in 2025 vs 876 in 2026 (Jan–Mar)

App & UI	Python · Streamlit · Altair charts · custom CSS · mobile- friendly
Logic	pandas · NumPy · rule-based matching, SDG mapping and momentum scoring
Live sources	arXiv cs.AI RSS via feedparser · Arbeitnow public API (no key) · JSON cache fallback
CV intake	pypdf · python-docx — text read in memory, never stored
Learning	18 matching rules: Kaggle Learn, Google MLCC, fast.ai, Hugging Face, DeepLearning.AI, Elements of AI, Coursera, Khan Academy · plus an HRD Corp e-LATiH pointer
Sponsor GenAI / cloud	Not used — no LLM by design, to prevent invented jobs or courses

Responsible AI: Transparent, Fair & Human-Centred

Human-centred

Never makes hiring, selection or promotion decisions. Results are for planning only.

No invention

Skills without a matching course are listed as “no resource” — never given a made-up one.

Every source shown

Dataset and posting count, arXiv and live ads with fetch time on each recommendation.

Privacy

CVs processed in memory for the session only. Public, open or synthetic data only.

Fairness check

Same skills tested across six backgrounds; background can shift a score by at most 10 points.

Resilient

If a live feed fails, cached data is used and labelled as cached.

Development Journey: From Problem to Prototype

Empathise

Started from the four user groups in the Domain 1 brief and their shared gap.

Define

One journey: profile → radar → ranked roles →pathway, each with a visible source.

Ideate

Rule-based, explainable scoring instead of a chatbot.

Prototype

Working Streamlit app on Kaggle, arXiv and Arbeitnow with cached fallbacks.

Test

Built-in fairness check now; Phase 2 usability tests with students and an HR / career-centre user.

Deploy

Public cloud URL + GitHub repo, low-bandwidth mode, 3 languages for mobile access.

04 · EXPECTED IMPACT

Impact framework: baseline → KPI → Phase 2 target

SDG TARGET

4.4

Increase youth and adults with relevant skills for employment and decent jobs.

Supporting:

· 4.3 lifelong learning

· 8.2 productivity through innovation

· 8.5 full and productive employment

BASELINE

30 pts

AI training gap: 56% use AI tools vs 26% formally trained. 600,000+ workers need reskilling in 3–5 years (MOSTI)

CORE KPI · PATHWAY COVERAGE

~39% → ≥ 80%

prototype today* — Phase 2 target

of skill gaps matched to a cited free or low-cost resource.

+ profile to pathway in < 5 min in user tests · 3 / 3 trend sources shown with source and date

*Our measurement of the current resource list: 36 of the dataset's 93 skills match a cited resource. Formal starting coverage is measured at the start of Phase 2.

Current Limitations & Phase 2 Roadmap

KNOWN LIMITATIONS

Small samples: 2026 covers only part of the year — the radar shows signals, not forecasts.

Live jobs aren’t Malaysian: Arbeitnow is mostly European listings, so its effect is small and bounded.

Keyword matching: arXiv and live-ad signals are rough indicators.

CV detection: recognises dataset skill names only; scanned PDFs can’t be read.

Rule-based mappings: SDG tags and the resource list are limited.

PHASE 2 ROADMAP

1. Deploy to a public cloud URL with the GitHub repository.
2. Expand learning resources toward ≥ 80% gap coverage.
3. Usability tests with students and an HR / career-centre user; fix the friction found.
4. Measure KPIs: coverage, time to pathway, sources shown.

Thank You

Navigate your skills. Discover your future.

BY TEAM CALADOR

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SDC Hackathon 2026 · Hosted by XMUM AI Club